

FUXIANG ZHAI

MPhil Student in Artificial Intelligence · HKUST (Guangzhou)
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RESEARCH INTERESTS

Image Generation · Unified Understanding & Generation · Latent Visual Reasoning · Reward Modeling & Evaluation · Agentic Generation

EDUCATION

The Hong Kong University of Science and Technology (Guangzhou)

2025 – Present

MPhil in Artificial Intelligence, Information Hub | Advisor: Prof. Lei Zhu

- Research on image generation, unified multimodal models, latent reasoning, and generative model evaluation.

Xinjiang University

2020 – 2024

B.Eng. in Computer Science and Technology, School of Intelligence Science and Technology (School of Future Technology)

- Excellent Graduate; ranked in the top 12% of the major.

PUBLICATIONS & PREPRINTS

Latent Action Control for Reasoning-Guided Unified Image Generation — *arXiv, under review* · First Author

Fuxiang Zhai, Sixiang Chen, Yingjin Li, Shuaibo Li, Jianyu Lai, Tengjun Huang, Lei Zhu | [arXiv:2605.16961](#)

GenEvolve: Self-Evolving Image Generation Agents via Tool-Orchestrated Visual Experience Distillation — *arXiv, under review*

Sixiang Chen, Zhaohu Xing, Tian Ye, Xinyu Geng, Yunlong Lin, Jianyu Lai, Xuanhua He, Fuxiang Zhai, Jialin Gao, Lei Zhu | [arXiv:2605.21605](#)

ROAD: Rule-Grounded Context-Aware Open-World Driver Anomaly Detection — *arXiv, under review*

Fuxiang Zhai et al. | [OpenReview: 9p9YEuoqnX](#)

PosterReward: Unlocking Accurate Evaluation for High-Quality Graphic Design Generation — *CVPR 2026*

Jianyu Lai*, Sixiang Chen*, Jialin Gao*, Hengyu Shi, Zhongying Liu, Fuxiang Zhai, Junfeng Luo, Xiaoming Wei, Lujia Wang, Lei Zhu | [arXiv:2603.29855](#)

RESEARCH EXPERIENCE

MPhil Researcher — Generative AI & Image Generation

2025 – Present

HKUST (Guangzhou), supervised by Prof. Lei Zhu

- Lead first-author research on Latent Action Control (LAC), turning planning, internal drafting, diagnosis, and refinement into hidden continuous actions for unified image generation.
- Investigate post-training and reward-driven optimization for generative models, including latent/flow reinforcement learning and evaluation on compositional and knowledge-grounded generation benchmarks.
- Contribute to GenEvolve, PosterReward, and ROAD, spanning agentic image generation, reward modeling for graphic design, and open-world anomaly detection.

SELECTED RESEARCH PROJECTS

Semi-Supervised Estrous Cycle Classification from Vaginal Cytology
Xinjiang University · National Undergraduate Innovation Training Project

- Developed a four-class medical image classification pipeline for Proestrus, Estrus, Metestrus, and Diestrus using labeled and unlabeled microscopy images.
- Explored domain adaptation, dual-classifier discrepancy learning, confident pseudo-labeling, and label-noise cleaning; achieved approximately 88% overall accuracy on a private dataset.
- Led dataset preprocessing, training, evaluation, ablation analysis, and reproducible experiment management.

Undergraduate

Reward Modeling & Evaluation for Generative Images
HKUST (Guangzhou)

- Built and evaluated reward-model pipelines using PosterReward, HPSv3, ImageReward, PickScore, and CLIP-based metrics.
- Developed batch scoring and benchmarking workflows for image-generation systems and preference-based evaluation.

2025 – 2026

Embodied Visual Data Augmentation
HKUST (Guangzhou)

- Studied frame-consistent image editing for robot manipulation sequences, preserving task geometry while varying appearance, lighting, and backgrounds.
- Experimented with Qwen-Image-Edit, Flux Kontext, OmniGen2, ControlNet/LoRA, and diffusion-based editing pipelines.

2025

SELECTED HONORS

- Kaggle Bronze Medal — RSNA 2024 Lumbar Spine Degenerative Classification, Top 6%.
- First Prize — 14th and 15th National University Students Mathematics Competition in Xinjiang.
- Autonomous Regional People's Government Encouragement Scholarship.
- Outstanding Graduate — Xinjiang University.
- Third Prize, National Finals — Huajiao Cup competition.
- Outstanding Student League Cadre; Outstanding Volunteer — Xinjiang University.

TECHNICAL SKILLS

Programming & ML: Python, PyTorch, Hugging Face Transformers/Diffusers, vLLM, OpenCV, NumPy, MATLAB.
Systems & Tools: Linux, Slurm, CUDA, Git, LaTeX, ComfyUI; experience with multi-GPU A800 clusters and large-model inference/training workflows.

ACADEMIC ACTIVITIES

- Invited participant, Chinese Mathematical Society Annual Meeting (2021).
- Former head of the Student Union Office; Youth League publicity representative; online teaching volunteer.